

2.Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

STEPS:

Step 1: Download Datafile

Step 2: Load data file → pandas

Step 3: Display the Data File

Step 4: Check for missing values

Step 5: Fill or drop missing values

Step 6: Summary statistics

Step 7: Group by product and calculate the sales & quantity

Step 8: Bar plot of Total sales / product

Step 9: Line plot of sales over time

Step 10: Pivot table to analyze sales by region and product

Step 11: Correlation matrix

Step 12: heatmap of correlation matrix

PROGRAM:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
data = {
```

```
    "S.No": [1, 2, 3, 4, 5],
```

```
    "Date": ["2026-08-10","2026-08-11","2026-08-12","2026-08-13","2026-08-14"],
```

```
    "Product Name": ["Headphones","Keyboard",None,"Monitor","Keyboard"],
```

```
    "Sales": [15000,22000,18000,None,25000],
```

```
    "Quantity": [8,None,6,4,10],
```

```
    "Region": ["Thanjavur","Erode","Vellore",None,"Tirunelveli"]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
df.to_csv("sales_data.csv", index=False)
```

```
print("Data file created successfully!")
```

```
print("File name: sales_data.csv")

#LOAD DATA FILE

df = pd.read_csv("sales_data.csv")

print("\n DATA FILE LOADED ")

display(df)

#DISPLAY THE DATA FILE

print("\n SALES DATA ")

display(df)

#CHECK FOR MISSING VALUES

print("\nMISSING VALUES ")

missing_values = df.isnull().sum()

print(missing_values)

#FILL MISSING VALUES

df["Sales"] = df["Sales"].fillna(df["Sales"].mean())

df["Quantity"] = df["Quantity"].fillna(df["Quantity"].mean())

df["Product Name"] = df["Product Name"].fillna("Unknown")

df["Region"] = df["Region"].fillna("Unknown")

print("\n DATA AFTER HANDLING MISSING VALUES ")

display(df)

# SUMMARY STATISTICS

print("\nSUMMARY STATISTICS ")

display(df.describe())

#GROUP BY PRODUCT

product_summary = df.groupby(

    "Product Name"

)[["Sales", "Quantity"]].sum()

print("\n SALES AND QUANTITY BY PRODUCT ")

display(product_summary)

#BAR PLOT

plt.figure(figsize=(8, 5))

product_summary["Sales"].plot(kind="bar")
```

```
plt.title("Total Sales by Product")

plt.xlabel("Product Name")

plt.ylabel("Total Sales")

plt.xticks(rotation=0)

plt.tight_layout()

plt.show()

#LINE PLOT

df["Date"] = pd.to_datetime(df["Date"])

daily_sales = df.groupby("Date")["Sales"].sum()

plt.figure(figsize=(8, 5))

plt.plot(

    daily_sales.index,

    daily_sales.values,

    marker="o"

)

plt.title("Sales Over Time")

plt.xlabel("Date")

plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()

#PIVOT TABLE

pivot_table = pd.pivot_table(

    df,

    values="Sales",

    index="Region",

    columns="Product Name",

    aggfunc="sum",

    fill_value=0

)

print("\n PIVOT TABLE ")
```

```
display(pivot_table)

#CORRELATION MATRIX

correlation = df[

    ["Sales", "Quantity"]

].corr()

print("\nCORRELATION MATRIX ")

display(correlation)

#HEATMAP

plt.figure(figsize=(6, 4))

sns.heatmap(

    correlation,

    annot=True,

    cmap="coolwarm",

    fmt=".2f"

)

plt.title("Correlation Heatmap")

plt.tight_layout()

plt.show()
```

OUTPUT:

Data file created successfully!
File name: sales_data.csv

DATA FILE LOADED

	S.No	Date	Product Name	Sales	Quantity	Region
0	1	2026-08-10	Headphones	15000.0	8.0	Thanjavur
1	2	2026-08-11	Keyboard	22000.0	NaN	Erode
2	3	2026-08-12	NaN	18000.0	6.0	Vellore
3	4	2026-08-13	Monitor	NaN	4.0	NaN
4	5	2026-08-14	Keyboard	25000.0	10.0	Tirunelveli

SALES DATA

	S.No	Date	Product Name	Sales	Quantity	Region
0	1	2026-08-10	Headphones	15000.0	8.0	Thanjavur
1	2	2026-08-11	Keyboard	22000.0	NaN	Erode
2	3	2026-08-12	NaN	18000.0	6.0	Vellore
3	4	2026-08-13	Monitor	NaN	4.0	NaN
4	5	2026-08-14	Keyboard	25000.0	10.0	Tirunelveli

MISSING VALUES

S.No	0
Date	0
Product Name	1
Sales	1
Quantity	1
Region	1
dtype: int64	

DATA AFTER HANDLING MISSING VALUES

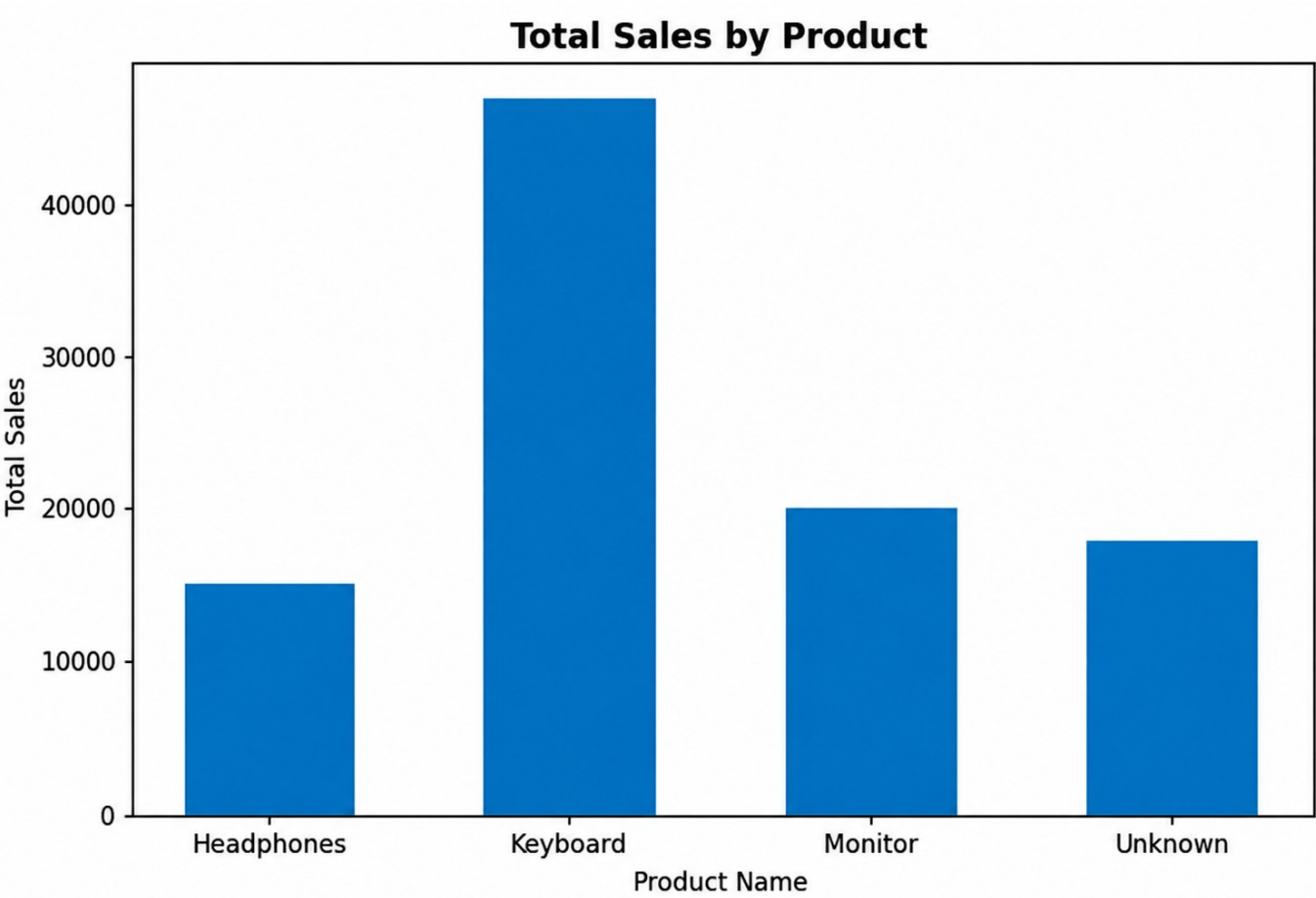
	S.No	Date	Product Name	Sales	Quantity	Region
0	1	2026-08-10	Headphones	15000.0	8.0	Thanjavur
1	2	2026-08-11	Keyboard	22000.0	7.0	Erode
2	3	2026-08-12	Unknown	18000.0	6.0	Vellore
3	4	2026-08-13	Monitor	20000.0	4.0	Unknown
4	5	2026-08-14	Keyboard	25000.0	10.0	Tirunelveli

SUMMARY STATISTICS

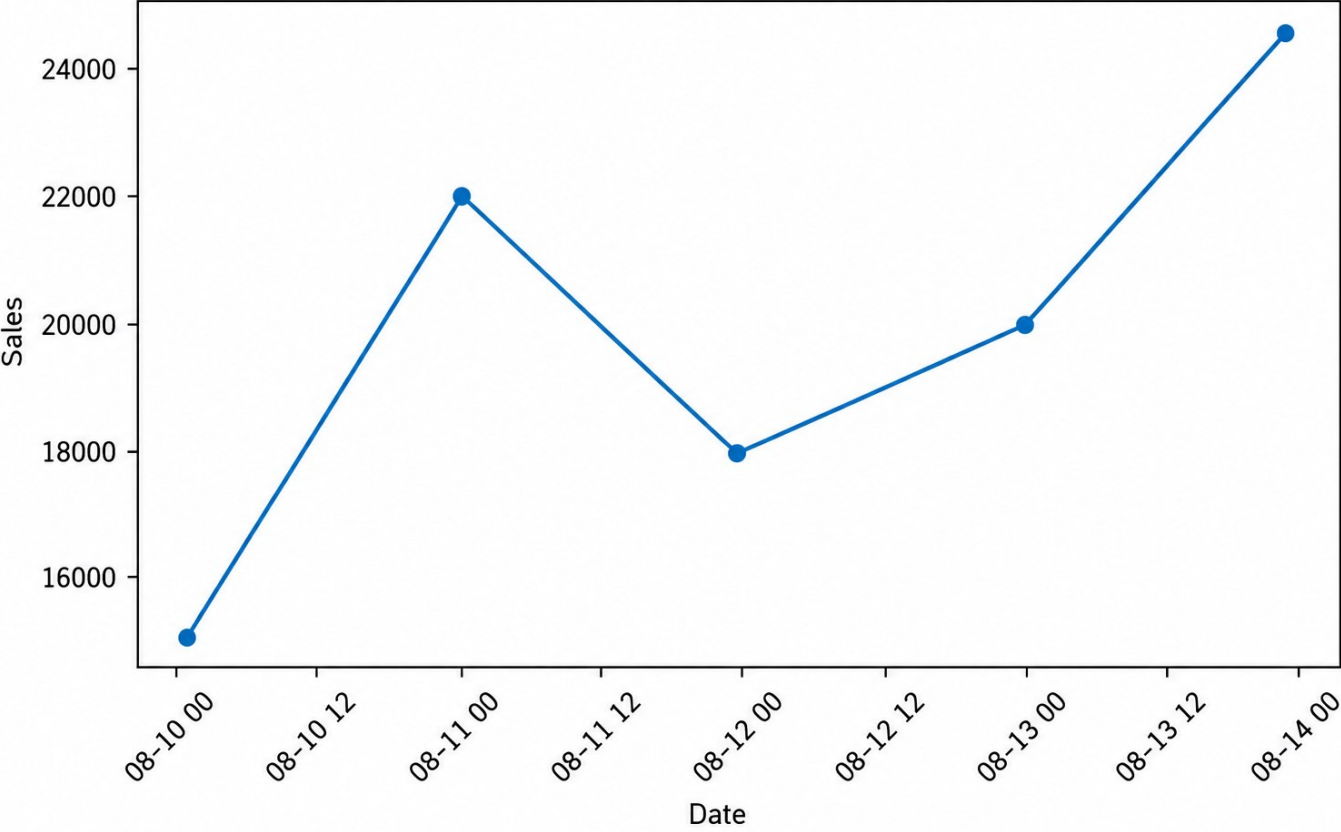
	S.No	Sales	Quantity
count	5.000000	5.000000	5.000000
mean	3.000000	20000.000000	7.000000
std	1.581139	3807.886553	2.236068
min	1.000000	15000.000000	4.000000
25%	2.000000	18000.000000	6.000000
50%	3.000000	20000.000000	7.000000
75%	4.000000	22000.000000	8.000000
max	5.000000	25000.000000	10.000000

SALES AND QUANTITY BY PRODUCT

Product Name	Sales	Quantity
Headphones	15000.0	8.0
Keyboard	47000.0	17.0
Monitor	20000.0	4.0
Unknown	18000.0	6.0



Sales Over Time



PIVOT TABLE

Product Name	Headphones	Keyboard	Monitor	Unknown
Region				
Erode	0.0	22000.0	0.0	0.0
Thanjavur	15000.0	0.0	0.0	0.0
Tirunelveli	0.0	25000.0	0.0	0.0
Unknown	0.0	0.0	20000.0	0.0
Vellore	0.0	0.0	0.0	18000.0

CORRELATION MATRIX

	Sales	Quantity
Sales	1.000000	0.352332
Quantity	0.352332	1.000000

